VICTORY 2027

SOUND

LECTURE-01 PHYSICS

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Topics To Be Covered

1. SOUND WAVES
2. RELATIONSHIP BETWEEN THE WAVELENGTH, WAVE VELOCITY AND FREQUENCY
3. SPEED OF SOUND IN DIFFERENT MEDIA
4. QUESTIONS


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SOUND WAVES

'Sound is a form of energy produced by rapidly vibrating objects. We hear sounds because this energy stimulates the auditory nerve in the human ear.'

- Our ears are sensitive only to a limited range of frequencies from 20 Hz to 20,000 Hz. This range of frequency is, therefore, called the **range of audibility**.
- However, the audibility range of a person decreases as he gets older since the hearing sensitivity of ears falls for both the low and high frequencies.
- The sound of frequency above 20, 000 Hz is called the **ultrasonic**, while the sound of frequency below 20 Hz is called the **infrasonic**.
- Both the ultrasonic and infrasonic are inaudible to human beings, but they both travel in a medium with the speed same as that of the audible sound.

$f < 20\ Hz$ Infrasonic Sound

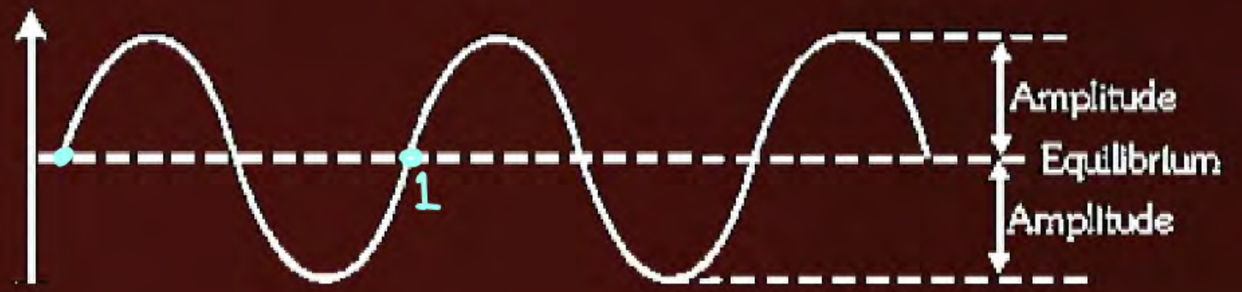
$20Hz < f < 20000\ Hz$ Audible Sound

$f > 20000\ Hz$ Ultrasonic Sound

$$V_{In} = V_{au} = V_{ul}$$

SOME TERMS RELATED TO WAVE MOTION

Amplitude When a wave passes through a medium, the maximum displacement of the particle of medium on either side of its mean position, is called the amplitude of wave.





SOME TERMS RELATED TO WAVE MOTION

Time period: The time taken by a particle of medium to complete it's one vibration is called the time period of wave. It is denoted by the letter T. Its S.I. unit is second (s).

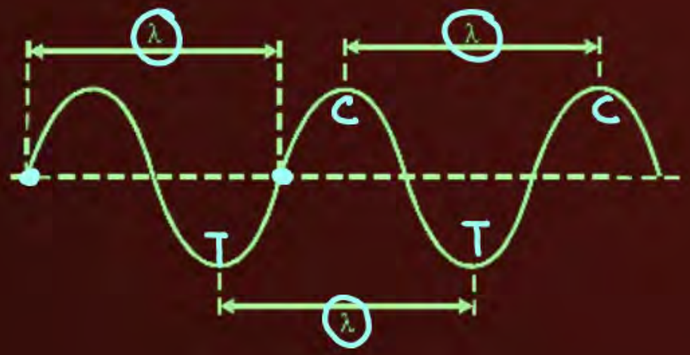
Frequency: The number of vibrations made by a particle of medium in one second is called the frequency of wave. It is same as the number of waves passing through a point in one second.

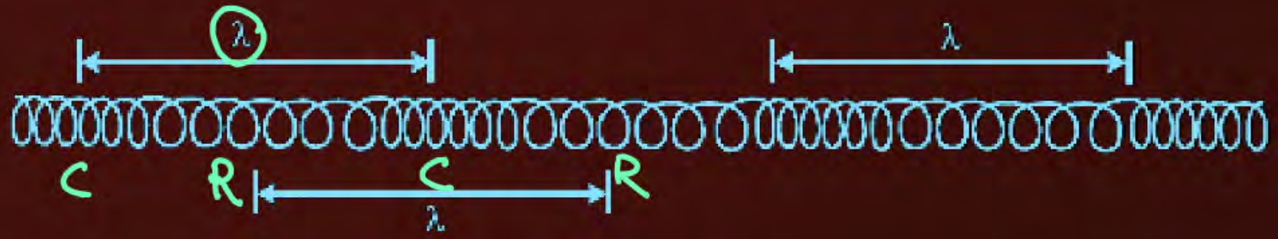
$$f = \frac{1}{T}$$

SOME TERMS RELATED TO WAVE MOTION

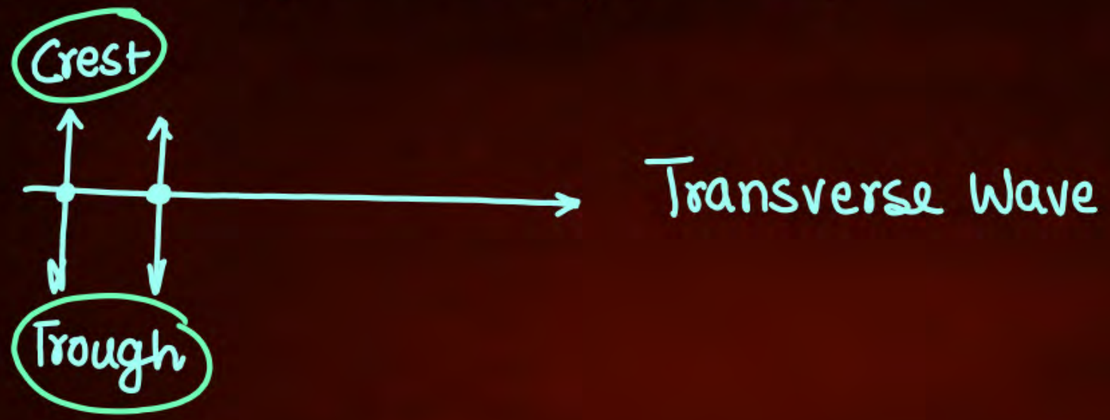
Wavelength: The distance travelled by the wave in one time period of vibration of particle of the medium, is called its wavelength.

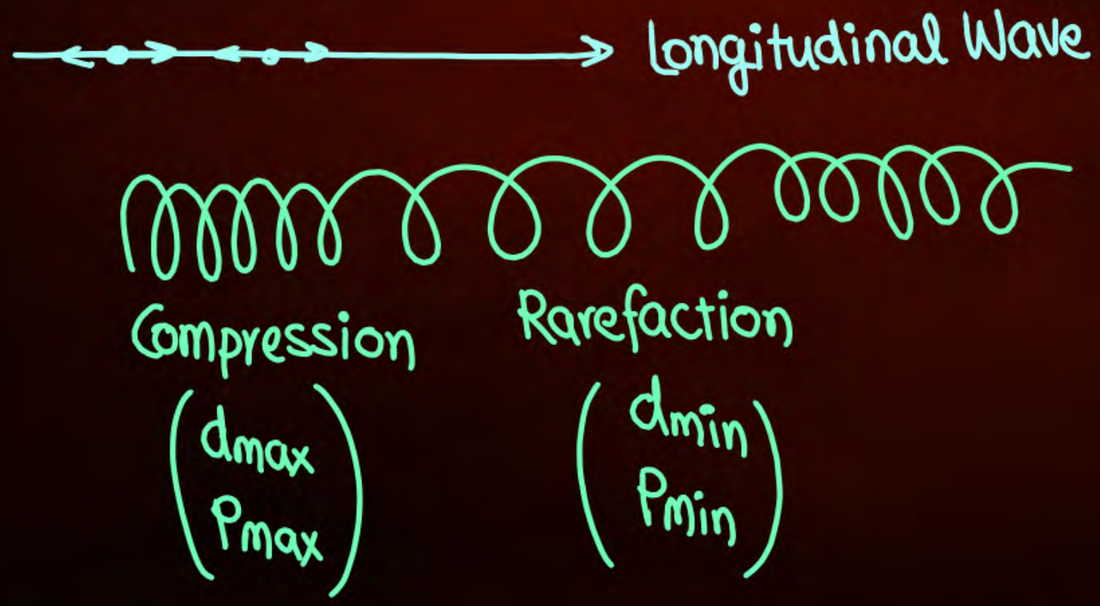
- It depends on the medium in which the wave travels.
- In a longitudinal wave, the distance between two consecutive compressions or between two consecutive rarefactions is equal to one wavelength,
- while in a transverse wave, the distance between two consecutive crests or between two consecutive troughs is equal to one wavelength.





(On Basis of motion of particles)





SOME TERMS RELATED TO WAVE MOTION

Wave velocity: The distance travelled by a wave in one second is called its wave velocity or wave speed.

- It may be noted that the wave velocity is constant for a given medium.
- It depends on the elasticity and the density of the medium.
- It changes when the wave passes from one medium to the other medium. (Refraction)

$$V = \frac{d}{t}$$

In one time period,

$$V = \frac{\lambda}{T}$$

$$\boxed{V = f\lambda}$$



RELATIONSHIP BETWEEN THE WAVELENGTH, WAVE VELOCITY AND FREQUENCY

Let velocity of a wave be V, time period T, frequency f and wavelength λ. By the definition of wavelength,

Wavelength λ = Distance travelled by the wave in one time period

Wavelength λ = **Wave velocity** $\times$ **Time period** $= V \times T$

$$VT = \lambda$$

But

$$T = \frac{1}{f}$$

$\therefore$ From equation

$$V \times \left(\frac{1}{f}\right) = \lambda$$

Or $V = f\lambda$

Wave velocity = Frequency × Wavelength



TYPES OF WAVE (On Basis of medium)

1

Mechanical Waves:

These waves require a medium for their propagation. For example, **sound waves** travel through the air to reach your ears. **Ocean waves** move through water to reach the shore.

A medium is the substance or material in which a wave will move. A medium can be a solid, a liquid, a gas, or a combination of these.

2

Non-mechanical waves:

These waves do not require any medium for their propagation. They can travel through space. These waves are also called **electromagnetic waves**. For example, light, radio waves, gamma rays, X-rays, ultraviolet rays, infrared rays, microwaves, all are electromagnetic waves.



Transverse Waves

A transverse wave is a wave where the movement of the particles of the medium is perpendicular to the direction of propagation of the wave.



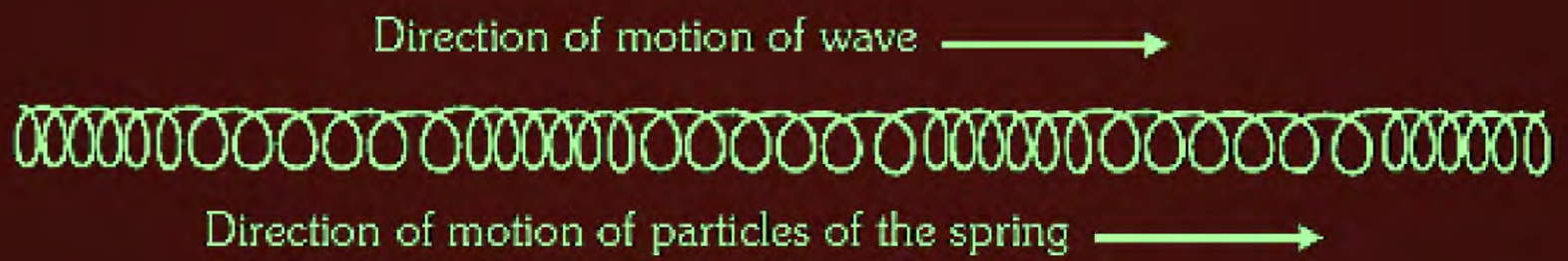
Transverse Waves

Crest and Trough: Transverse waves consist of moving crests and troughs. A crest is the part of the transverse wave which is above the line of zero disturbance. A trough is the part of transverse wave which is below the line of zero disturbance.

Longitudinal Waves

A longitudinal wave is a wave where the particles in the medium move parallel to the direction of propagation of the wave.

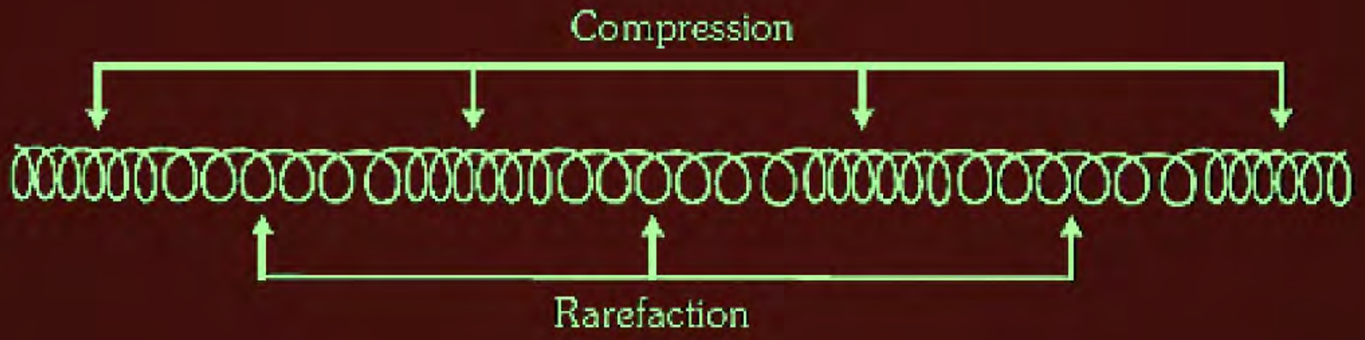
For example: If a long spring is given jerk along its length, a longitudinal wave is produced in it. Sound is also a longitudinal wave.



Longitudinal Waves

Compression (or condensation): A compression is a region in a longitudinal wave where the particles are closer together than their normal separations.

Rarefaction: A rarefaction is a region in a longitudinal wave where the particles are farther apart than their normal separations.



Speed of Sound

→ elasticity

→ density

$$v_s = \sqrt{\frac{E}{d}}$$

According to Newton,

T = Constant

(Isothermal Process)

$$v_s = \sqrt{\frac{E_{isothermal}}{d}}$$

$$v_s = \sqrt{\frac{P}{d}}$$

$$v_s = \sqrt{\frac{1.01 \times 10^5}{1.29}} \approx 280 \text{ m/s} \quad \times$$

According to Laplace,

T ≠ Constant

(Adiabatic Process) ⇒ No exchange of heat

$$v_s = \sqrt{\frac{E_{adiabatic}}{d}}$$

$$v_s = \sqrt{\frac{\gamma P}{d}}$$

$$\gamma = \frac{C_p}{C_v}$$

C_p → Specific Heat at Constant Pressure

C_v → Specific Heat at Constant volume

$\gamma_{air} \approx 1.4$

$$v_s = \sqrt{\frac{1.4 \times 1.01 \times 10^5}{1.29}}$$

$v_s \approx 332$ m/s ✓

SPEED OF SOUND IN DIFFERENT MEDIA

The speed of sound in a medium is given by the relation

$$V = \sqrt{\frac{E}{\rho}}$$

where E is the modulus of elasticity (Young's modulus in case of solids or bulk modulus in case of fluids) and ρ is the density of the medium.

Newton assumed that when sound travels in a gas, the temperature of gas does not change (i.e., the propagation of sound is an isothermal change). For isothermal change, modulus of elasticity is equal to the pressure i.e., $E = P$. Thus from eqn., speed of sound in a gas is given as

$$\boxed{V = \sqrt{\frac{P}{\rho}}}$$

SPEED OF SOUND IN DIFFERENT MEDIA

Later on, the scientist Laplace applied a correction to the above relation. According to **Laplace**, when sound travels in a gas, during the formation of compression and rarefaction, there is no exchange of heat in the medium i.e., the propagation of sound is an adiabatic change. For an adiabatic change, modulus of elasticity $\mathbf{E = \gamma P}$ where γ is the ratio of the specific heat at constant pressure to the specific heat at constant volume $(\gamma = C_p/C_v)$ and P is the pressure of the gas.

Hence from eqn., the speed of sound in a gas is given as

$$\mathbf{V = \sqrt{\frac{\gamma P}{\rho}}}$$

The value of γ depends on the nature of the medium. For air, γ is 1.4.

FACTORS AFFECTING THE SPEED OF SOUND IN A GAS

The speed of sound in a gas is affected by the change in

(i) Density
(ii) Temperature
(iii) Humidity, and
(iv) Direction of wind.

$$v_s = \sqrt{\frac{\gamma P}{d}}$$

(i) Effect of density: From relation $V = \sqrt{\frac{\gamma P}{\rho}}$, it is clear that $V \propto \frac{1}{\sqrt{\rho}}$ i.e., the speed of sound is inversely proportional to the square root of density of the gas.

FACTORS AFFECTING THE SPEED OF SOUND IN A GAS

(ii) **Effect of temperature:** The speed of sound in a gas increases with the increase in temperature of the gas. The reason is that with the increase in temperature, the density of gas decreases and consequently the speed of sound increases. In fact, the speed of sound is directly proportional to the square root of temperature of the medium i.e., $V \propto \sqrt{T}$ where T is the temperature of the gas on the Kelvin scale.

The speed of sound in air increases by about $0.61 ms^{-1}$ for each degree celsius rise in temperature (provided that the rise in temperature is not very large). i.e.,

$$V_t = V_0 + 0.61t$$

at $0^\circ C \Rightarrow V_0 = V_0 + 0.61 \times 0 = 332$ m/s

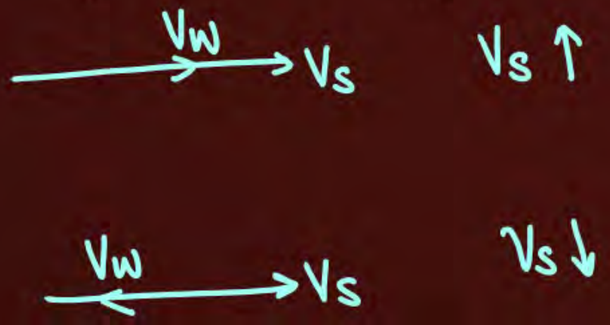
at $25^\circ C \Rightarrow V_t = 332 + 0.61 \times 25 = 347.25$ m/s.

$$V_t = 332 + 0.61 \times t$$

FACTORS AFFECTING THE SPEED OF SOUND IN A GAS

(iii) **Effect of humidity:** The speed of sound in air increases with the increase in humidity in air. The density of water vapour is about 5/8 times the density of dry air at ordinary temperature, therefore the increase of moisture in air tends to decrease the density of air. Hence the **speed of sound in the humid air is greater than the speed of sound in dry air**. In other words, the sound travels faster in humid air than in dry air.

(iv) **Effect of direction of wind:** The speed of sound increases or decreases according to the direction of travel of wind. If wind is blowing in the direction of propagation of sound, the speed of sound increases, while if it is blowing in a direction opposite to that of sound, the speed of sound decreases.



DISTINCTION BETWEEN LIGHT AND SOUND WAVES

HW

Light waves	Sound waves
1. These are the electro-magnetic waves.	1. These are the mechanical waves.
2. They can travel in vacuum	2. They require a material medium for propagation.
3. The speed of light waves is very high ($= 3 \times 10^8 ms^{-1}$ in air).	3. The speed of sound waves is low ($= 330 ms^{-1}$ in air).
4. The wavelength of light waves (visible) is very small, of the order of $10^{-6} m$.	4. The wavelength of sound waves is in the range of $10^{-2} m$ to 10 m .
5. These waves are transverse.	5. In air, these waves are longitudinal.

QUESTIONS

The diagram below shows
(a) displacement-time, and
(b) displacement-distance, graph of a wave travelling in a string with velocity 20 ms^{-1}. In each case, use graph to calculate the frequency and wavelength of the wave.

$V = 20\ m/s$

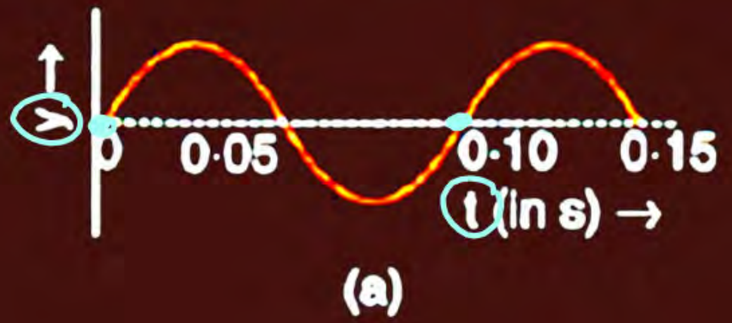
(a) $T = 0.10\ s.$

$f = \frac{1}{T} = \frac{1}{0.10} = 10\ Hz.$

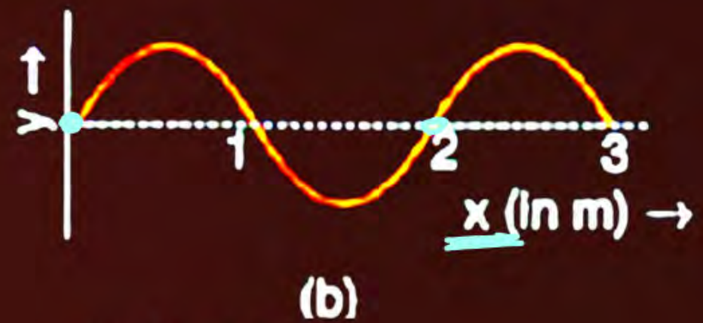
$V = f\lambda$

$20 = 10\lambda$

$\boxed{\lambda = 2m}$



(a)



(b)

$V = 20\ m/s$

$\lambda = 2m$

$V = f\lambda$

$20 = f(2)$

$f = 10Hz.$



QUESTIONS

To detect the obstacles in their path, bats produce:

1. infrasonic waves
2. ultrasonic waves
3. electromagnetic waves
4. radio waves

FACTORS NOT AFFECTING THE SPEED OF SOUND IN A GAS

The speed of sound in a gas is not affected by the change in (i) pressure, (ii) amplitude of wave, and (iii) wavelength or frequency of wave.

Imp

(i) **Effect of pressure:** The speed of sound in a gas is independent of pressure. In the formula $V = \sqrt{\frac{\gamma P}{\rho}}$, the ratio $\frac{P}{\rho}$ remains unchanged with the change in pressure. When pressure increases, volume decreases, but mass remains unchanged, so density increases, such that the ratio P/ρ remains constant.

$$V_s = \sqrt{\frac{\gamma P}{d}}$$

P ↑ d ↑

P ↓ d ↓

$\frac{P}{d}$ = Constant



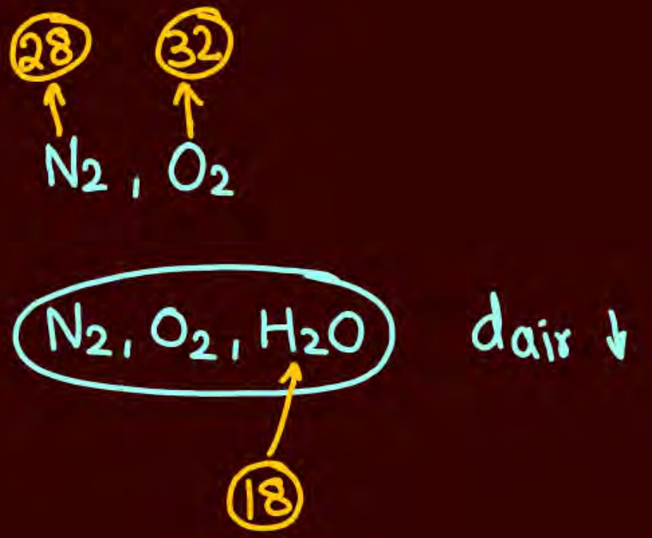
FACTORS NOT AFFECTING THE SPEED OF SOUND IN A GAS

(ii) **Effect of amplitude of wave:** The speed of sound does not depend on the amplitude of sound wave.

(iii) **Effect of wavelength (or frequency) of wave:** The speed of sound does not depend on the wavelength (or frequency) of sound wave.

humidity ↑ d_{air} ↓ v_s ↑

humidity ↓ d_{air} ↑ v_s ↓



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You